Non-fermentative gram-negative bacteria (NFB)

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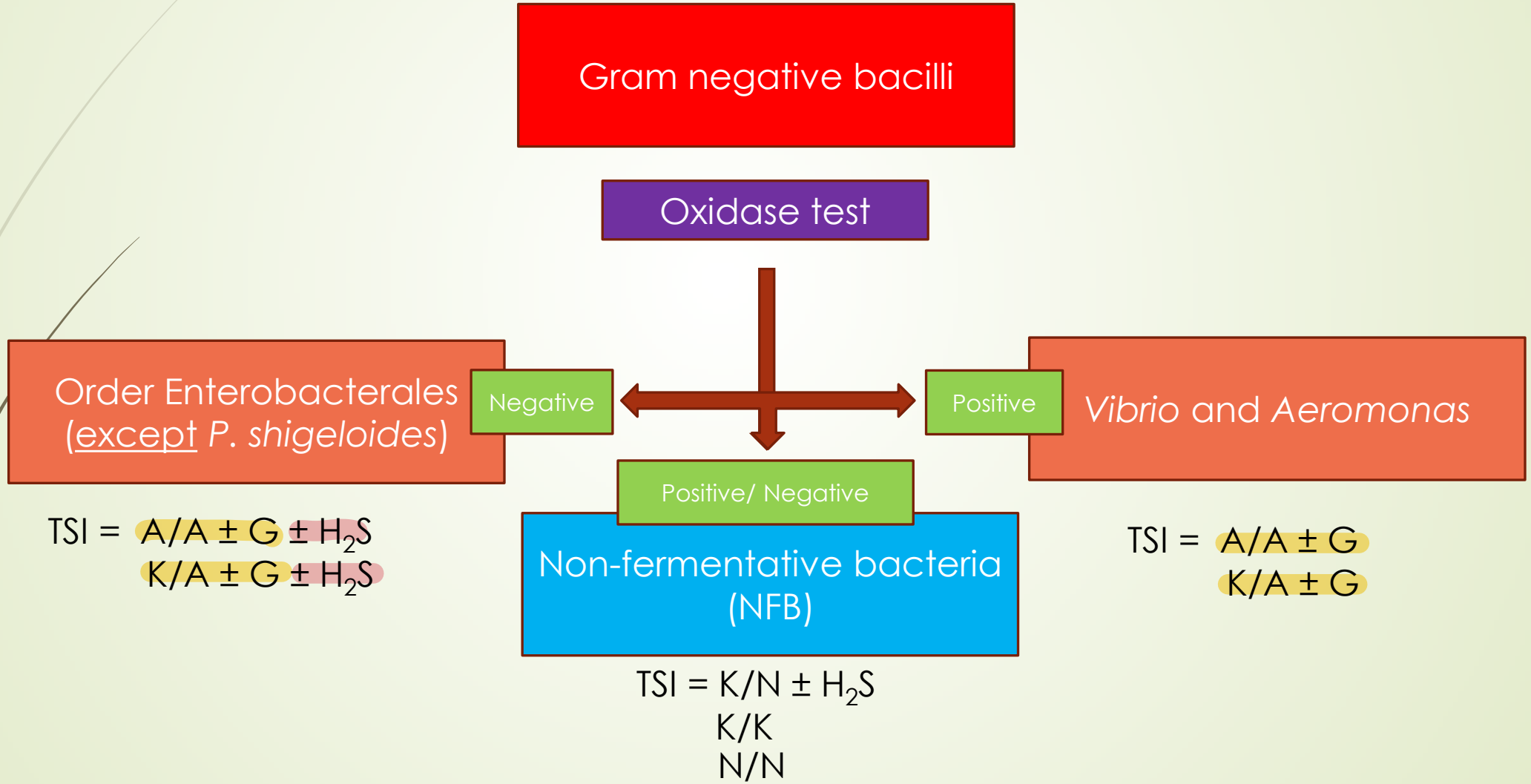
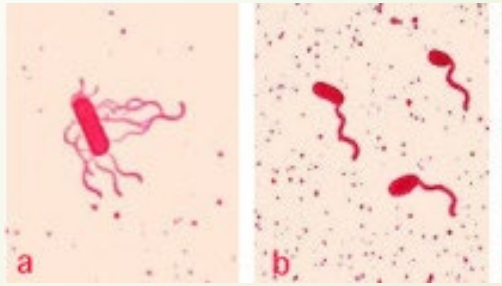
Review on gram negative bacilli



General characteristic of NFB (1)

- Gram negative bacteria (bacilli, coccobacilli, diplococci)
- Non-glucose fermenter
  - Oxidizer e.g. *Acinetobacter baumannii*, *Burkholderia pseudomallei*
  - Asaccharolytic bacteria e.g. *Acinetobacter lwoffii, Moraxella catarrhalis*
- Catalase: mostly positive
- Oxidase: positive or negative
- Flagella: - peritrichous flagella (a)
  - polar flagella (b)

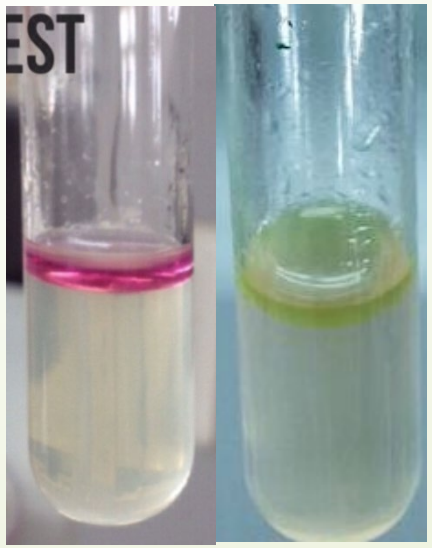


General characteristic of NFB (2)

- Usually found in environment e.g.
  - soil
  - Water
  - plants or vegetable
- Opportunistic infection
- Nosocomial infection
- Some strains can be identified as true pathogen e.g.
  - *Burkholderia pseudomallei*
  - *Elizabethkingia meningoseptica*
- Some strains can cause both human or plant diseases e.g.
  - *Burkholderia cepacia*

Biochemical test for identification of NFB

- Biochemical test
  - Glucose O/F: +/- or -/-
  - TSI: K/N, K/K, K/N H_2S, A/N
    - oxidize น้ำตาลได้
    - ไม่ได้ น้ำตาลเลย
  - Motility
  - Indole
  - Citrate
  - Urease test
  - Nitrate reduction test
  - Nitrite reduction test



Biochemical test for identification of NFB

- Biochemical test (addition)
  - DNase test
  - Gelatin hydrolysis ดูว่าสร้าง Enzyme gelatinase ได้
  - Starch hydrolysis
  - Esculin hydrolysis
  - Acetamide utilization
  - Sugar utilization (oxidization)
  - Growth at 42°C (TSA medium)
  - Pigment production (King P/ King F)
  - Cetrimide tolerance test

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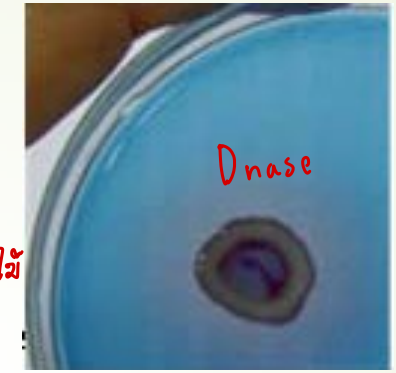

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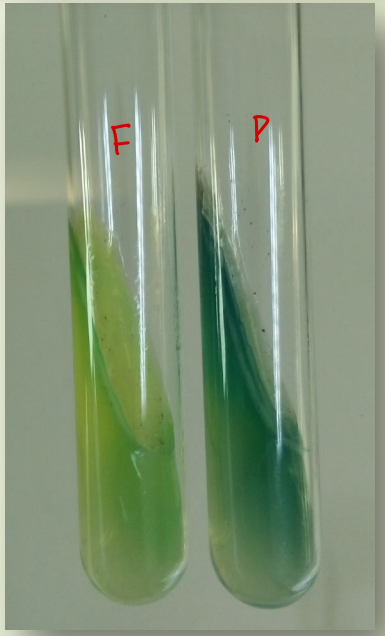

Classification of NFB

- Based on rRNA-DNA hybridization and 16S rRNA sequencing
  - rRNA homology group I : ***Pseudomonas***
  - rRNA homology group II : ***Burkholderia***, ***Ralstonia***, ***Cupriavidus***, *Pandoraea*
  - rRNA homology group III : *Delftia, Comamonas, Acidovorax*
  - rRNA homology group IV : *Brevundimonas*
  - rRNA homology group V : ***Stenotrophomonas***
  - Ungroup in group I-V : ***Shewanella***, *Sphingomonas, Methylobacterium, Roseomonas*

    ทนต่อเกลือ

Other group of NFB

- *Acinetobacter* and related bacteria : ***Acinetobacter***
- *Moraxella* and related bacteria : ***Moraxella***, *Neisseria*, *Oligella*, *Psychobacter*
- *Alcaligenes* and related bacteria *:* ***Alcaligenes****, Achromobacter*
- *Flavobacterium* and related bacteria *:* ***Flavobacterium, Chryseobacterium****,* ***Elizabethkingia****,* ***Empedobacter****,* ***Myroides****, Sphingobacterium, Pedobacter, Weeksella, Bergeyella, Balnaetrix*

Pseudomonas spp. classification

(rRNA homology group I)

1. Fluorescent Group	2. Non-fluorescent Group
- ***Pseudomonas aeruginosa*** - ***Pseudomonas fluorescens*** - ***Pseudomonas putida*** - *Pseudomonas veronii* - *Pseudomonas monteilii* - *Pseudomonas mosselii*	**2.1 Stutzeri Group** - *Pseudomonas stutzeri* - *Pseudomonas mendocina* **2.2 Alcaligenes Group** - *Pseudomonas alcaligenes* - *Pseudomonas pseudoalcaligenes* - *Pseudomonas* spp. CDC group 1

Pseudomonas spp. classification

(rRNA homology group I)

1. Fluorescent Group
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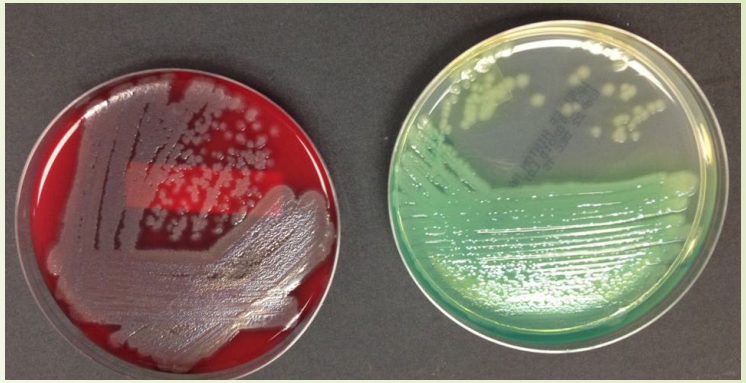
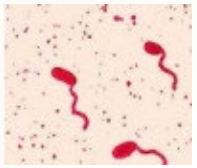
Pseudomonas aeruginosa (1)

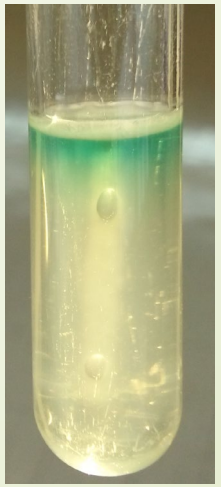


- ***Pseudomonas aeruginosa*** is most common strain of **fluorescens group**
- Grape-like odor กลิ่นองุ่น
- Oxidase: positive
- Motile with polar flagella
- Growth on TSA at 42°C
- King P/ King F: positive/ positive
- Nitrate reduction test ($NO_3 \rightarrow NO_2 \rightarrow \rightarrow \rightarrow N_2$)
- Utilization of monosaccharide not disaccharide
- Utilization of acetamide as carbon source



* ใช้ acetamide เป็น carbon source *

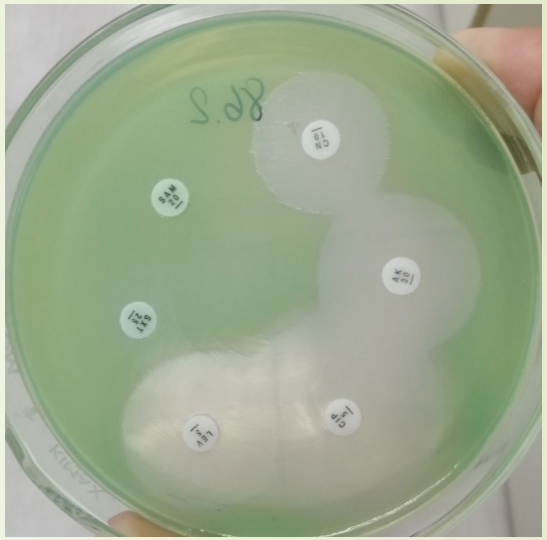
motile

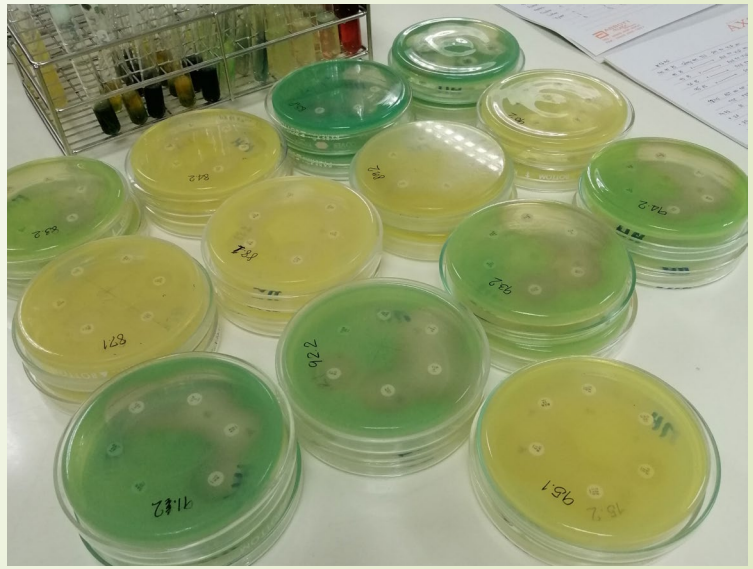


Pseudomonas aeruginosa (2)

- Pigment production
  - Pyocyanin (Blue-green pigment) → King P (Medium A)
  - Pyoverdin (Green-yellow pigment) → King F (Medium B)
  - Pyorubin (Red pigment)
  - Pyomelanin (Brown pigment)
  - Apyocyanogenic strain (no-pigment)
- * 1st causative agent of nosocomial infection *
- Multidrug resistant (MDR) bacteria

ดื้อยา 3 กลุ่ม ขึ้นไป !





Pseudomonas aeruginosa (3)

- Virulent factors of *P. aeruginosa*
  - Fimbria (Pili) → attachment to the epithelial cell
  - Alginate polysaccharide (capsule and biofilm production) → anti-phagocyte, anti-drug
    คล้าย LPS ทำให้เกิด mucoid
  - Extracellular enzyme e.g. protease, elastase → anti-human immune respond
  - Pyocyanin, Pyoverdin → Siderophore (iron chelator) แย่งจับ เหล็ก
  - Hemolysin (Phospholipase C) and lecithinase → Cell membrane destruction
  - Cytotoxin (leukocidin) → WBC destruction
  - Endotoxin (LPS) → Sepsis

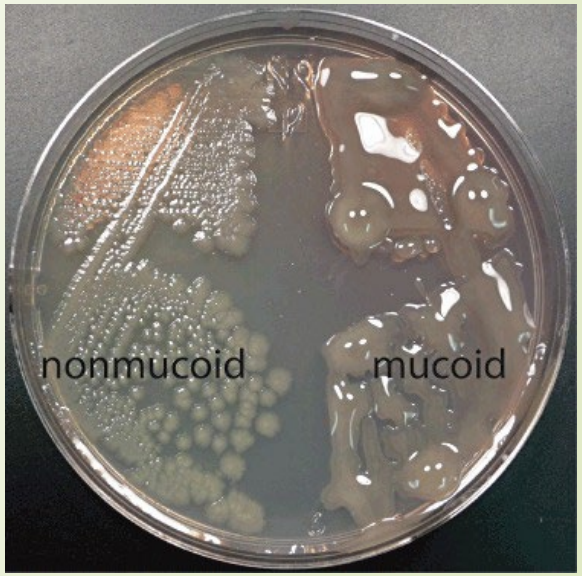
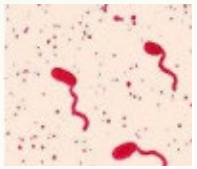


Photo credit : www.researchgate.net/publication/224008_Pr391

Pseudomonas fluorescens & P. putida

- ***P. fluorescens*** and ***P. putida*** were categorized in **fluorescens group**
- Oxidase: positive
- Motile with polar flagella



- Produce only pyoverdin pigment, but not pyocyanin pigment

Only Pyoverdin : King F (+)

- ! King P/ King F: negative/ positive !
- ! Do not growth on TSA at 42°C (differentiated from *P. aeruginosa*)
- ! Do not reduce nitrate (differentiated from *P. aeruginosa*)
- ! Utilization of xylose 100%
- ***P. fluorescens*** (+) can be differentiated from ***P. putida*** (-) by gelatin hydrolysis

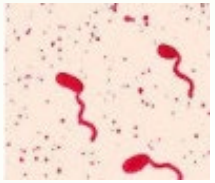
Pseudomonas spp. classification

(rRNA homology group I)

2. Non-fluorescent Group
2.1 Stutzeri Group - *Pseudomonas stutzeri* - *Pseudomonas mendocina* **2.2 Alcaligenes Group** - *Pseudomonas alcaligenes* - *Pseudomonas pseudoalcaligenes* - *Pseudomonas* spp. CDC group 1

Stutzeri group:
Pseudomonas stutzeri

- ***P. stutzeri*** is most common strain of **non-fluorescens group**
- Oxidase: positive
- Motile with polar flagella

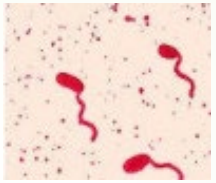


- ! Dry and Wrinkled colony (48 hours)
- Growth on TSA at 42°C
- ! Hydrolyze starch 100% (+)
- No production of arginine dihydrolase (AD) AD: (-)

Photo credit : A. Kesara Wataksorn

Stutzeri group:
Pseudomonas mendocina

- ***P. mendocina*** is categorized in **non-fluorescens group**
- Oxidase: positive
- Motile with polar flagella

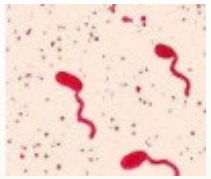


- ! Brownish and yellow, Smooth and flat colony
- Growth on TSA at 42°C
- ! Reduce nitrate to nitrogen gas ($NO_3 \rightarrow NO_2 \rightarrow \rightarrow \rightarrow N_2$)
- ! Production of arginine dihydrolase (AD) AD : +
- Cannot hydrolyze starch and gelatin

  − −

Alcaligenes group:
Pseudomonas alcaligenes & P. pseudoalcaligenes

- ***P. alcaligenes*** and ***P. pseudoalcaligenes*** are categorized in **non-fluorescens group**
- Rarely isolated in clinical specimen
- Biochemical inert strains
- Oxidase: positive
- Motile with polar flagella



- ***P. alcaligenes*** cannot oxidized fructose while ***P. pseudoalcaligenes*** can do.

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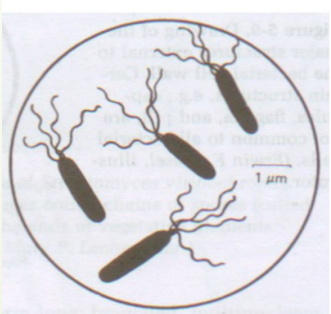
Burkholderia pseudomallei (1)

Photo credit: en.wikipedia.org/wiki/*Burkholderia_pseudomallei*

- Wrinkle colony (48 hour)
- Earth smelling odor กลิ่นดิน
- Pale-pink colony on MacConkey agar (lactose oxidizer in 48 hours)
- Selective media: **Ashdown's medium** (violet colony)
- Gram stain: GNB with Bipolar staining
- Lophotrichous (polar tufts) flagella

มีสีชมพูระเรื่อ : Lactose Oxidizer



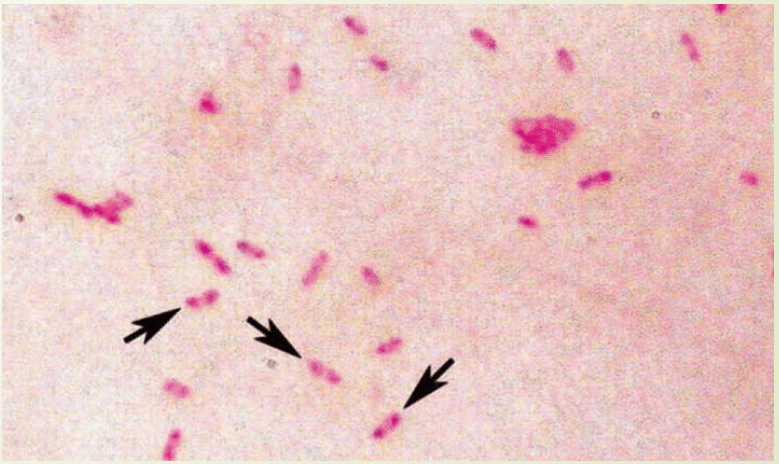
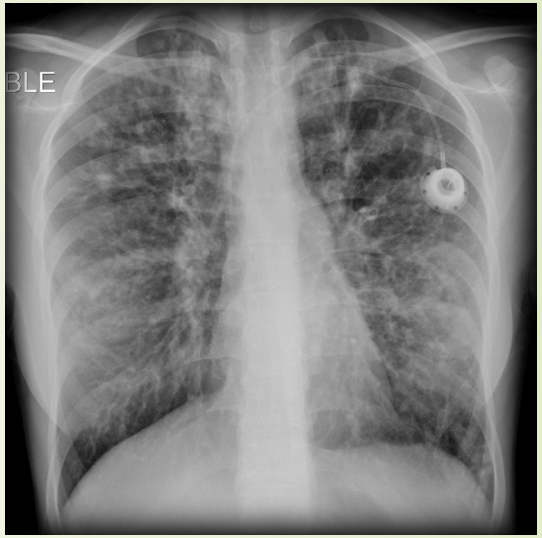


Photo credit: www.researchgate.net/publication/246629396

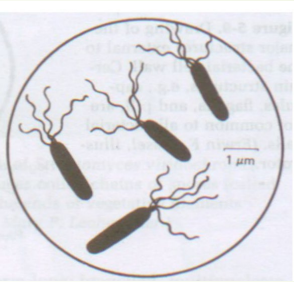
Burkholderia pseudomallei (2)

- Airborne infection (Risk factor group : level 3)
- Usually found in soil or mud
- Melioidosis disease: endemic area is Southeast Asia and Northern Australia
- Most found to cause disease in agriculture person (esp. farmers)
- Growth on TSA at 42°C
- Starch hydrolysis: negative (differentiated from *P. stutzeri*)
- Nitrate reduction test ($NO_3 \rightarrow N_2$) (differentiated from *B. cepacia* complex)
- Arabinose negative (differentiated from *B. thailandensis* Ara+)

Photo credit: step1.medbullets.com/respiratory/121999/cystic-fibrosis



Burkholderia cepacia complex (BCC)



- Oxidase: positive
- Usually produce yellow pigment colony
- Lophotrichous (polar tufts) flagella
- Found in cystic fibrosis (CF) and chronic granulomatous disease (CGD)
- Selective media: BCSA (*B. cepacia* selective agar), PCM (*P. cepacia* medium)
- Growth on TSA at 42°C
- No N_2 gas from nitrate reduction (differentiated from *B. pseudomallei*)

Photo credit: step1.medbullets.com/respiratory/121999/cystic-fibrosis

Ralstonia spp. & *Cupriavidus* spp.

- Found in moist environment: soil, river and lake
- *Ralstonia pickettii* is the most common in this group.
- Several former *Ralstonia* spp. are currently assigned to genus *Cupriavidus*
- *R. pickettii* is easily confused with *P. fluorescens* and *B. cepacia* (phenotypes)
- *R. pickettii* was first isolated from cystic fibrosis (CF) patients.
- Oxidase: positive
- Motile with Polar flagella
- ! ~~Growth~~ on TSA at 42°C !

No growth

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  - rRNA homology group I : ***Pseudomonas***
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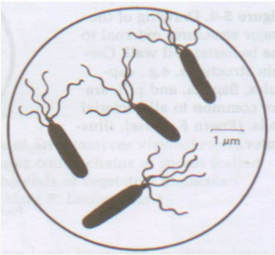
  - rRNA homology group III : *Delftia, Comamonas, Acidovorax*
  - rRNA homology group IV : *Brevundimonas*
  - rRNA homology group V : ***Stenotrophomonas***
  - Ungroup in group I-V : ***Shewanella***, *Sphingomonas, Methylobacterium, Roseomonas*

Stenotrophomonas maltophilia

(rRNA homology group V)

* nosocomial infection *

- Formerly name: *Xanthomonas (Pseudomonas) maltophilia*
  - Oxidase: negative ตัวแรกของสไลด์นี้
  - Motile: positive with polar tufts (differentiated from *Acinetobacter* spp.)
  - Lysine decarboxylation: positive
  - Esculin/ Gelatin hydrolysis: positive
  - DNase: positive
  - Maltose: positive (strong and rapid)
  - 3rd causative agent of nosocomial infection
  - Resistant to Carbapenems (β-lactam group)
  - Mostly sensitive to trimethoprim-sulfamethoxazole (SXT)



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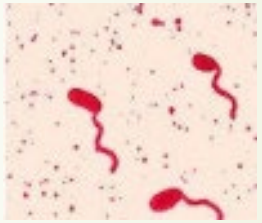
Shewanella spp.

(Ungroup in group I-V)

Photo credit : A. Kesara Wataksorn

- ***Shewanella putrefaciens*** and ***Shewanella algae***
  - Oxidase: positive
  - TSI: **K/N H_2S** (only one genus in NFB) มีแค่ตัวเดียวในเรื่องนี้
  - Motile: positive (polar flagella)
  - *S. algae* can grow on 6.5% NaCl (halophilic NFB), but *S. putrefaciens* cannot.
  - *S. algae* can grow on 42°C, but *S. putrefaciens* cannot .
  - *S. putrefaciens* can utilize sucrose and maltose, but *S. algae* cannot.
  - *S. algae* is most frequently isolated from clinical specimens, whereas *S. putrefaciens* is usually found in environmental source.



Other group of NFB

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- *Moraxella* and related bacteria : ***Moraxella***, *Neisseria*, *Oligella*, *Psychobacter*
- *Alcaligenes* and related bacteria *:* ***Alcaligenes****, Achromobacter*
- *Flavobacterium* and related bacteria *:* ***Flavobacterium, Chryseobacterium****,* ***Elizabethkingia****, Empedobacter, Myroides, Sphingobacterium, Pedobacter, Weeksella, Bergeyella, Balnaetrix*

Photo credit: ASM MicrobeLibrary.org

Acinetobacter spp.

- ***Acinetobacter baumannii*** is a most common strain
  - Pale-pink colony in MacConkey agar (lactose oxidizer)
  - Gram negative coccobacilli (short rod)
  - Oxidase: negative * Oxidase ⊖ ตัวที่ 2 ของสไลด์นี้
  - Non-motile
  - 2nd causative agent of nosocomial infection
  - Multidrug resistant (MDR) bacteria
  - Extensively drug resistant (XDR) bacteria
- ***Acinetobacter lwoffii***
  - Asaccharolytic bacteria

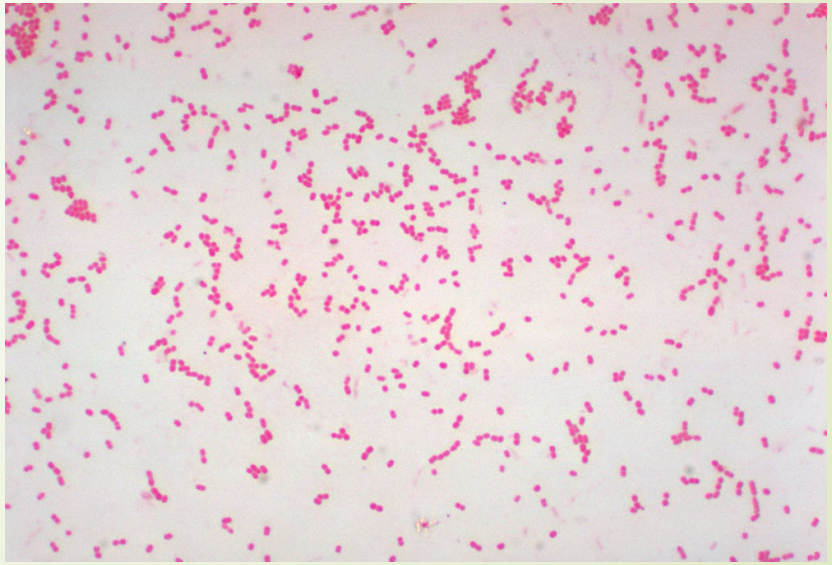


Photo credit: www.luciacangussu.bio.br/atlas/acinetobacter-spp/

Moraxella spp.

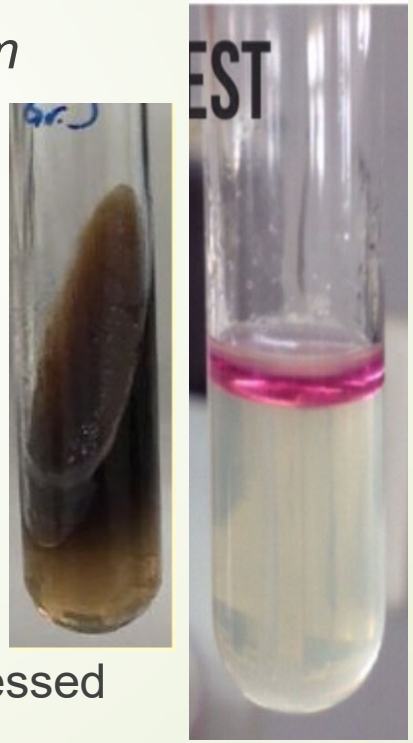
- Gram negative diplococci (*Moraxella* spp.→ gram negative short rod)
- Oxidase: positive
- No growth on MacConkey agar
- ***M. nonliquefaciens*** is most common isolated from clinical specimens.
- ***M. catarrhalis*** is the pathogen in respiratory tract, but it is normal flora in children.
- ***M. lacunata*** is associated with conjunctivitis.
- ***M. osloensis*** is normal flora in genital tract (non-fastidious bacteria)
- Non-motile
- Nitrate reduction test: positive
- Asaccharolytic bacteria

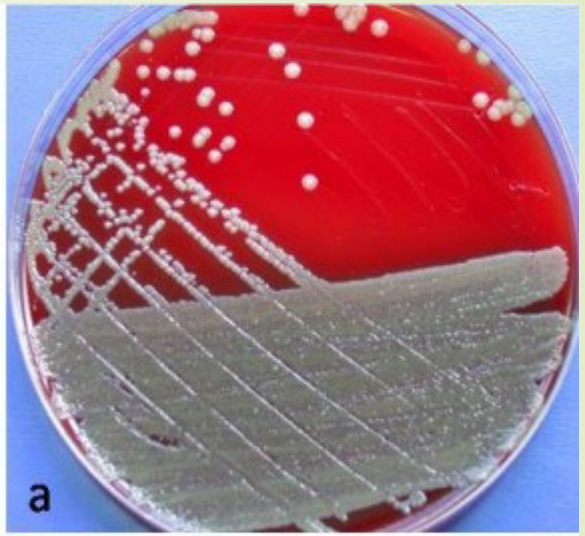
Alcaligenes spp.

- ***Alcaligenes faecalis*** is most common isolated from specimens.
  - Gram negative short rod
  - Oxidase: positive
  - Motile: positive with peritrichous flagella
  - Asaccharolytic bacteria
  - Reduce nitrite, but not reduce nitrate
  - Opportunistic infection

Elizabethkingia meningoseptica

- Formerly name: *Chryseobacterium meningosepticum*
  Flavobacterium meningosepticum
  - Yellow pigment colony
  - Gram negative slender rod
  - Oxidase: positive
  - **Indole: positive** Key
  - Non-motile
  - Esculin hydrolysis: positive (100%)
  - True pathogen of **neonatal meningitis**
  - Septicemia, pneumonia in adult and Immune suppressed





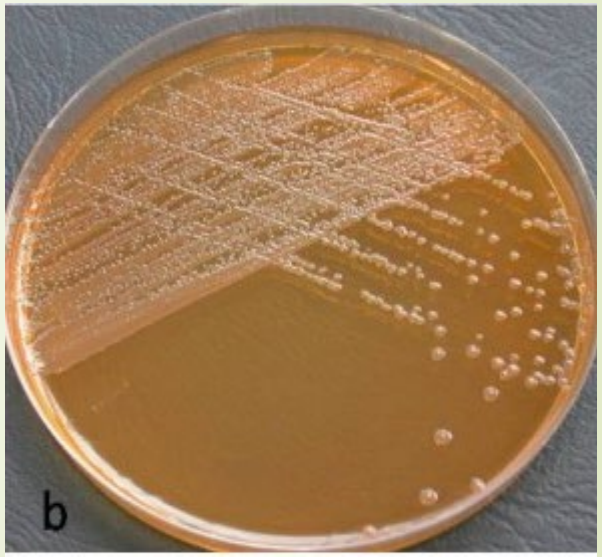


Photo credit : A. Kesara Wataksorn

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Biochemical test	*Pseudomonas aeruginosa*	*Pseudomonas fluorescens*	*Pseudomonas putida*	*Pseudomonas stutzeri*	*Pseudomonas mendocina*	*Pseudomonas alcaligenes*	*Pseudomonas pseudoalcaligenes*	*Burkholderia cepacia*	*Burkholderia pseudomallei*	*Ralstonia pickettii*	*Delftia acidovorans*	*Comamonas testosteroni*	*Brevundimonas diminuta*	*Brevundimonas vesicularis*	*Sphingomonas paucimobilis*	*Shewanella putrefaciens*	*Shewanella algae*	*Achromobacter xylosoxidants*	*Achromobacter denitricans*	*Alcaligenes faecalis*	*Oligella ureolytica*	*Bordetella bronchiseptica*	*Elizabethkingia meningoceptica*	*Myroides odoratus*	*Bergeyella zoohelcum*	*Weeksella virosa*	*Moraxella sp.*	*Acinetobacter lwoffii*	*Acinetobacter baumannii*	*Stenotrophomonas maltophilia*
Oxidase	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+/-	+	+	+	+	+	+	+	+	+	+/-	+	+	-	-	-
Growth on McA	+	+	+	+	+	+	+	+	+	+/-	+	+	+	+/-	-	+	+	+	+	+	+/-	+	-	-/+	-	-	-	+	+	+
Pigment	B,F,G	F	F	B,Y	-	-	-	Y	-	-	-	-	-	B,Y	Y	B	-	-	-	-	-	-	Y	Y	B	B	-	-	-	Y
H_2S in TSI	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	+	+	-	-	-	-	-	-	-	-	-	-	-	-	-
Motility	+	+	+	+	+	+	+/-	+	+	+	+	+	+	+	+/-	+	+	+	+	+	+	+	-	-	-	-	-	-	-	+
Flagella	1	>1	>1	1	1	1	1	>1	>1	1	>1	>1	1	1	1	1	1	P	P	P	P	P	-	-	-	-	-	-	-	>1
Indole	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	+	+	+/-	+	-	-	-	-
King P	+/-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
King F	+/-	+/-	+/-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Urease	+/-	+	+	+/-	+	+/-	-	+	-	+	-	-	-/+	-	-	-	+	-	-	-	+	+	-/+	+	+	-	-	-	-/+	-
Nitrate	+/-	-	-	+	+	+/-	+	-	+	+	+/-	+	-	-/+	-	+	+	+	+	-/+	+/-	+	-	-	-	-	V	-	-	-/+
N_2 gas	-/+	-	-	+	+	-	-	-	+	+/-	-	-	-	-	-	-	-	+/-	+	-	-	-	-	-	-	-	-	-	-	-
Growth 42°C	+	-	-	+	+	+/-	+	-/+	--	-	-	-/+	+/-	-	-/+	-	+	+	+/-	+/-	-	+	-	-	+/-	+	V	-/+	+	-/+
Esculin	-	-	-	-	-	-	-	+	+	-	-	-	-	+	+	-	-	-	-	-	-	-	+	-	-	-	-	-	-	+
DNase	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-/+	+	+	-	-	-	-	-	+	+	-	+/-	-	-	-	+
ONPG	-	-	-	-	-	-	-	+	-	-	-	-	-	+/-	+	-	-	-	-	-	-	-	+	-	-	-	-	-	-	-/+
Gelatin	+/-	+	-	-	-	-	-	+	+	-/+	-	-	-/+	-	-	+/-	+	-	-	-	-	-	+	+	+/-	+	-	-	-	+
Starch	-/+	+/-	-/+	+	-	-	-	-	+	-/+	-	-	-	+/-	+/-	-	-	-	-	-	-	-	-/+	-	-	-	-	-	-	-
6.5% NaCl	+/-	+/-	+/-	+	+	-/+	+	-	-	-/+	-	-	-/+	-/+	-	-/+	+	+/-	+/-	+	-/+	+/-	-	+/-	-	-	-	-/+	-/+	-/+
Glucose	+	+	+	+	+	-	-/+	+	+	+	-	-	-/+	+/-	+	+/-	-	+	-	-	-	-	+	-	-	-	-	-	+	+
Xylose	+	+	+	+	+	-	-	+	+	+	-	-	-	-	+	-	-	+	-	-	-	-	-	-	-	-	-	-	+	-/+
Mannitol	+/-	+	-/+	+	-/+	-	-	+/-	+	-	+	-	-	-	-	-	-	-	-	-	-	-	+	-	-	-	-	-	-	-
Lactose	-	-	-	-	-	-	-	+	+	+	-	-	-	-	+	-	-	-	-	-	-	-	-/+	-	-	-	-	-	+	-/+
Maltose	-	-	-	+	-	-	-	+	+	+	-	-	-	-/+	+	-/+	-	-	-	-	-	-	+	-	-	-	-	-	-/+	+
Fructose	+	+	+	+	+	-	+	+/-	+	+	+	-	-	-	+	-/+	-	-	-	-	-	-	+	-	-	-	-	-	-	+
Sucrose	-	-/+	-	-	-	-	-	+	-	-	-	-	-	-	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-/+

THE END

ANY QUESTIONS ?